



INSTITUTO POLITÉCNICO NACIONAL
UNIDAD PROFESIONAL INTERDISCIPLINARIA DE
INGENIERÍA Y TECNOLOGÍAS AVANZADAS



CARRERA: INGENIERÍA TELEMÁTICA

UNIDAD DE APRENDIZAJE: MULTIMEDIA

PRACTICA: 9 MP3



ALUMNO:

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GRUPO: 2TV5

-PARA EL PRIMER AUDIO:

```
from mutagen.mp3 import MP3
from mutagen.id3 import ID3, TIT2, TPE1, TALB, TCON, TYER, TRCK, COMM, APIC
import os

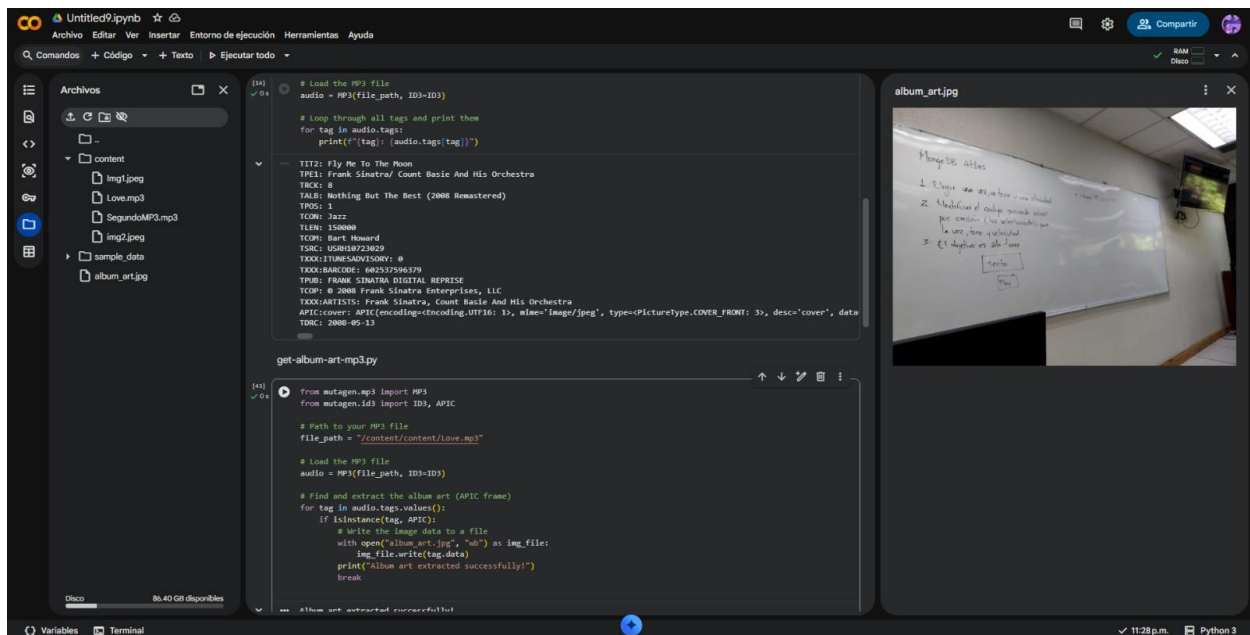
path_mp3 = "/content/content/Love.mp3"
path_imagen = "/content/content/Img1.jpeg" # Imagen nueva

if os.path.exists(path_mp3):
    try:
        audio = ID3(path_mp3)
    except:
        audio = ID3()

    # Aplicar imagen
    if os.path.exists(path_imagen):
        with open(path_imagen, 'rb') as img_file:
            audio.delall('APIC') # Borrar anteriores
            audio.add(APIC(
                encoding=3,
                mime='image/jpeg',
                type=3,
                desc='Cover',
                data=img_file.read()
            ))
            print("Imagen añadida correctamente.")

        audio.save(path_mp3)
        print("Metadatos actualizados con éxito!")
    else:
        print("No encontré el archivo MP3 en la carpeta 'mp3/'.")

... Metadatos actualizados con éxito!
```



-PARA EL SEGUNDO AUDIO:

The screenshot shows a code editor with a file explorer on the left, a terminal at the bottom, and a video window on the right. The file explorer shows a directory structure with files like 'img1.jpeg', 'Love.mp3', 'SegundoMP3.mp3', 'img2.jpeg', 'sample_data', and 'album_art.jpg'. The code editor displays two Python scripts: 'get-album-art-mp3.py' and 'read-mp3-metadata.py'. The 'get-album-art-mp3.py' script extracts album art from an MP3 file and saves it as 'album_art.jpg'. The 'read-mp3-metadata.py' script installs mutagen and reads metadata from an MP3 file. The terminal shows the output of the 'get-album-art-mp3.py' script, indicating that the album art was successfully extracted. The video window shows a person pointing at a whiteboard with diagrams.

```
TSRC: USRH10723029
TXXX:ITUNESADVISORY: 0
TXXX:ISRC: 002537596379
TPUB: FRANK SINATRA DIGITAL REPRISE
TCOP:  2008 Frank Sinatra Enterprises, LLC
TXXX:ARTIST: Frank Sinatra, Count Basie And His Orchestra
APIC:cover: APIC(encoding=<encoding.UTF16: 1>, mime='image/jpeg', type=PictureType.COVER_FRONT: 3, desc='cover', data
TDR: 2008-05-13

get-album-art-mp3.py

[45] ✓ 0s
from mutagen.mp3 import MP3
from mutagen.id3 import ID3, APIC

# Path to your MP3 file
file_path = "/content/content/SegundoMP3.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3-ID3)

# Find and extract the album art (APIC frame)
for tag in audio.tags.values():
    if isinstance(tag, APIC):
        # Write the image data to a file
        with open("album_art.jpg", "wb") as img_file:
            img_file.write(tag.data)
        print("Album art extracted successfully!")
        break

Album art extracted successfully!

read-mp3-metadata.py

[10] ✓ 4s
pip install mutagen

from mutagen.mp3 import MP3
from mutagen.id3 import ID3, TIT2, TPE1, TALB

# Path to your MP3 file
file_path = "/content/content/SegundoMP3.mp3"

# Load the MP3 file
```

Cambiar la caratula

```
from mutagen.mp3 import MP3
from mutagen.id3 import ID3, TIT2, TPE1, TALB, TCON, TYER, TRCK, COMM, APIC
import os

path_mp3 = "/content/content/SegundoMP3.mp3"
path_imagen = "/content/content/Img2.jpeg" # Imagen nueva

if os.path.exists(path_mp3):
    try:
        audio = ID3(path_mp3)
    except:
        audio = ID3()

    # Aplicar imagen
    if os.path.exists(path_imagen):
        with open(path_imagen, 'rb') as img_file:
            audio.delall('APIC') # Borrar anteriores
            audio.add(APIC(
                encoding=3,
                mime='image/jpeg',
                type=3,
                desc='Cover',
                data=img_file.read()
            ))
            print("Imagen añadida correctamente.")

        audio.save(path_mp3)
        print("Metadatos actualizados con éxito!")
    else:
        print("No encontré el archivo MP3 en la carpeta 'mp3/'.")
```

Metadatos actualizados con éxito!

-CODIGOS:

```
!pip install mutagen
```

cambio de la data:

```
from mutagen.id3 import ID3, TIT2, TPE1, TALB
from mutagen.mp3 import MP3
```

```
# Path to your MP3 file
file_path = "/content/content/SegundoMP3.mp3"
```

```
# Load the MP3 file
audio = MP3(file_path, ID3=ID3)
```

```
# Modify metadata (e.g., Title, Artist, Album)
audio.tags.add(TIT2(encoding=3, text="Vaije a la luna")) # Title
audio.tags.add(TPE1(encoding=3, text="Juanito alimaña")) # Artist
audio.tags.add(TALB(encoding=3, text="Las poderosas")) # Album
```

```
# Save changes
audio.save()
```

```
print("Metadata updated successfully!")
```

obtener metadata:

```
from mutagen.mp3 import MP3
from mutagen.id3 import ID3
```

```
# Path to your MP3 file
file_path = "/content/content/SegundoMP3.mp3"
```

```
# Load the MP3 file
audio = MP3(file_path, ID3=ID3)
```

```
# Loop through all tags and print them
for tag in audio.tags:
    print(f"{tag}: {audio.tags[tag]}")
```

obtener caratula:

```
from mutagen.mp3 import MP3
from mutagen.id3 import ID3, APIC
```

```
# Path to your MP3 file
file_path = "/content/content/SegundoMP3.mp3"
```

```
# Load the MP3 file
audio = MP3(file_path, ID3=ID3)
```

```
# Find and extract the album art (APIC frame)
for tag in audio.tags.values():
    if isinstance(tag, APIC):
        # Write the image data to a file
        with open("album_art.jpg", "wb") as img_file:
            img_file.write(tag.data)
        print("Album art extracted successfully!")
        break
```

lectura metadata:

```
!pip install mutagen
```

```
from mutagen.mp3 import MP3
```

```

from mutagen.id3 import ID3, TIT2, TPE1, TALB

# Path to your MP3 file
file_path = "/content/content/SegundoMP3.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3=ID3)

# Extract duration (in seconds)
duration = audio.info.length

# Extract ID3 tags (metadata)
tags = audio.tags

# Extract metadata if available
title = tags.get('TIT2', 'Unknown Title').text[0] # Song title
artist = tags.get('TPE1', 'Unknown Artist').text[0] # Artist name
album = tags.get('TALB', 'Unknown Album').text[0] # Album name

# Print metadata
print(f"Title: {title}")
print(f"Artist: {artist}")
print(f"Album: {album}")
print(f"Duration: {duration:.2f} seconds")

cambio de caratula:

from mutagen.mp3 import MP3
from mutagen.id3 import ID3, TIT2, TPE1, TALB, TCON, TYER, TRCK, COMM, APIC
import os

path_mp3 = "/content/content/SegundoMP3.mp3"
path_imagen = "/content/content/Img2.jpeg" # Imagen nueva

if os.path.exists(path_mp3):
    try:
        audio = ID3(path_mp3)
    except:
        audio = ID3()

    # Aplicar imagen
    if os.path.exists(path_imagen):
        with open(path_imagen, 'rb') as img_file:
            audio.delall('APIC') # Borrar anteriores
            audio.add(APIC(
                encoding=3,
                mime='image/jpeg',
                type=3,
                desc='Cover',
                data=img_file.read()
            ))
        print("Imagen añadida correctamente.")

    audio.save(path_mp3)
    print("Metadatos actualizados con éxito!")
else:
    print("No encontré el archivo MP3 en la carpeta 'mp3/'.")

```